PDEs and Complex Analysis

Based on lectures by Michele Della Morte

Notes taken by Mikkel Kielstrup Holst

These notes are not endorsed by the lecturers. All errors are mine.

Contents

1. Introduction

Everywhere in physics we describe phenomena by how something changes and typically relate this change to the quantity itself. This means differential equations pop up all over the place. Some examples include the Einstein field equations

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu}$$

and the Euler-Lagrange equations

$$\frac{\partial \mathcal{L}}{\partial x^\mu} - \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = 0$$

both of which look deceptively simple.

Since differential equations are everywhere we need to be able to solve them. This typically requires computing some very difficult integrals and many are simply impossible. However using techniques from complex analysis some become very easy. These tools are therefore introduced in the latter half of these notes.

2. Ordinary Differential Equations

An *ordinary differential equation* of order n is an equation of the form

$$f\left(x, y, \frac{\mathrm{d}y}{\mathrm{d}x}, ..., \frac{\mathrm{d}^n y}{\mathrm{d}x^n}\right) = 0$$

with $y = y(x)$. We typically care about *linear* differential equations in physics. These are of the form

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = \sum_{i=0}^{n-1} a_i(x) \frac{\mathrm{d}^i y}{\mathrm{d}x^i} + \underbrace{r(x)}_{\text{source}}$$

with $a_i(x)$ and $r(x)$ being continuous. When $r(x) = 0$ we call the equation *homogeneous*, these always have the trivial solution $y(x) = 0$.

2.1. Separation of variables

Consider an equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = F(x, y)$$

Assuming $F(x, y) = f(x)g(y)$ we can integrate the above giving

$$\int f(x) \,\mathrm{d}x = \int \frac{1}{g(y)} \,\mathrm{d}y$$

This is called *separation of variables*.

Example 2.1. Consider

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x(1 + y)$$

We find

$$\int x \,\mathrm{d}x = \int \frac{\mathrm{d}y}{1 + y}$$

$$\Rightarrow y = Ae^{x^2/2} - 1$$

with A being an arbitrary constant.

2.2. Exact equations

We can write

$$F(x,y) = -\frac{A(x,y)}{B(x,y)}$$

Then

$$A\,\mathrm{d}x + B\,\mathrm{d}y = 0$$

This differential is *exact* if

$$A = \frac{\partial U}{\partial x}; \quad B = \frac{\partial U}{\partial y}$$

Meaning we have

$$\mathrm{d}U \equiv \frac{\partial U}{\partial x}\,\mathrm{d}x + \frac{\partial U}{\partial y}\,\mathrm{d}y \overset{!}{=} A\,\mathrm{d}x + B\,\mathrm{d}y$$

For some function $U(x,y)$. This is the case if

$$\frac{\partial A}{\partial y} = \frac{\partial B}{\partial x}$$

Then $\mathrm{d}U = 0$ implying $U(x,y) = k$. We integrate the equation for A to find

$$U = \int A\,\mathrm{d}x + F(y)$$

To determine $F(y)$ we use the equation for B giving

$$B = \frac{\partial U}{\partial y}$$

$$= \int \frac{\partial A}{\partial y}\,\mathrm{d}x + \frac{\mathrm{d}F}{\mathrm{d}y}$$

which is quite nice.

Example 2.2. Consider

$$x\frac{\mathrm{d}y}{\mathrm{d}x} + 3x + y = 0$$

We find

$$A = 3x + y; \quad B = x$$

We have

$$\frac{\partial A}{\partial y} = \frac{\partial B}{\partial x}$$

so it is exact. Then

$$U = \int 3x + y \,\mathrm{d}x + F(y)$$

$$= \frac{3}{2}x^2 + xy + k_1 + F(y) \overset{!}{=} k_2$$

or

$$\frac{3}{2}x^2 + xy + F(y) = k$$

We also have

$$\underbrace{B}_{x} = \int \mathrm{d}x + \frac{\mathrm{d}F}{\mathrm{d}y}$$

$$= x + k_1 + \frac{\mathrm{d}F}{\mathrm{d}y}$$

$$\overset{!}{=} \frac{\partial U}{\partial y} = x + \underbrace{\frac{\mathrm{d}F}{\mathrm{d}y}}_{\overset{!}{=}0}$$

so $k_1 = 0$ and

$$F(y) = k_2$$

Then

$$\frac{3}{2}x^2 + xy = k$$

and we are done.

2.3. Inexact equations

We now assume the equation is *inexact* meaning

$$A \,\mathrm{d}x + B \,\mathrm{d}y = 0; \quad \frac{\partial A}{\partial y} \neq \frac{\partial B}{\partial x}$$

We will try to make this exact by finding an *integrating factor* $\mu(x, y)$ such that

$$\frac{\partial(\mu A)}{\partial y} = \frac{\partial(\mu B)}{\partial x}$$

This works since multiplying the differential by μ does not change it. Assume $\mu = \mu(x)$ then

$$\mu\frac{\partial A}{\partial y} = \mu\frac{\partial B}{\partial x} + B\frac{\mathrm{d}\mu}{\mathrm{d}x}$$

or

$$\frac{\mathrm{d}\mu}{\mu} = f(x)\,\mathrm{d}x$$

with

$$f(x) \equiv \frac{1}{B}\left(\frac{\partial A}{\partial y} - \frac{\partial B}{\partial x}\right)$$

We can integrate this to obtain

$$\mu(x) = \exp\left[\int f(x)\,\mathrm{d}x\right]$$

Alternatively with $\mu = \mu(y)$ we find

$$\mu(y) = \exp\left[\int g(y)\,\mathrm{d}y\right]$$

with

$$g(y) \equiv \frac{1}{A}\left(\frac{\partial B}{\partial x} - \frac{\partial A}{\partial y}\right)$$

Any arbitrary constant in the above has been thrown away since we just need some μ that works.

Example 2.3. Consider

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2}{y} + \frac{3y}{2x} = 0$$

where

$$A = 4x + 3y^2; \quad B = 2xy$$

This is clearly inexact. We find

$$f(x) = \frac{2}{x}$$

giving

$$\mu(x) = x^2$$

Then we multiply the differential by x^2 giving

$$\left(4x^3 + 3x^2y^2\right) \mathrm{d}x + 2x^3y \, \mathrm{d}y = 0$$

which is exact by construction. We proceed as before

$$U = \int A \, \mathrm{d}x + F(y)$$

$$= x^4 + x^3y^2 + k_1 + F(y) \stackrel{!}{=} k_2$$

so

$$x^4 + x^3y^2 + F(y) = k$$

And

$$\underbrace{B}_{2x^3y} = \int \frac{\partial A}{\partial y} \, \mathrm{d}x + \frac{\mathrm{d}F}{\mathrm{d}y}$$

$$= \int 6x^2y \, \mathrm{d}x + \frac{\mathrm{d}F}{\mathrm{d}y}$$

$$= 2x^3y + k_1 + \frac{\mathrm{d}F}{\mathrm{d}y}$$

$$\stackrel{!}{=} \frac{\partial U}{\partial y} = 2x^3y + \underbrace{\frac{\mathrm{d}F}{\mathrm{d}y}}_{0}$$

so $k_1 = 0$ and

$$F(y) = k_2$$

Then

$$x^4 + x^3y^2 = k$$

and we are done.

We now consider a special type of inexact ordinary differential equation. Namely those of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

which is a first order linear ordinary differential equation. We have

$$A = P(x)y - Q(x); \quad B = 1$$

We find

$$\mu(x) = \exp\left[\int P(x)\,\mathrm{d}x\right]$$

After making the differential exact we find

$$Q(x)\mu(x) = \mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + \mu(x)P(x)y$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y)$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu(x)y) = \mu(x)\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)\mu(x)y$$

We can then integrate to find

$$y = \frac{1}{\mu(x)}\int Q(x)\mu(x)\,\mathrm{d}x$$

Example 2.4. Consider

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy = 4x$$

We find

$$\mu(x) = e^{x^2}$$

Then

$$
\begin{aligned}
y &= e^{-x^2} \int e^{x^2} 4x \,\mathrm{d}x \\
&= 2e^{-x^2} \int e^u \,\mathrm{d}u \\
&= 2e^{-x^2} \left(e^{x^2} + k_1 \right) \\
&= ke^{-x^2} + 2
\end{aligned}
$$

and we are done.

3. Partial Differential Equations

Consider a higher dimensional space X with $X \subseteq \mathbb{R}^n$. A *partial differential equation* relates a function $u : X \to \mathbb{R}$ or $\mathbb{C}$ which depends on n independent variables $x_1, ..., x_n$ to its partial derivatives. The computation of u typically involves a boundary value problem leading to a dependence on the geometry of X as well as the values of u and its derivatives on the boundary ∂X.

Example 3.1. The following are all examples of *homogeneous* and *linear* partial differential equations we find in physics:

$$\text{The transport equation:} \quad a\frac{\partial u}{\partial x} + b\frac{\partial u}{\partial y} = 0$$

$$\text{The Laplace equation:} \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\text{The Schrödinger equation:} \quad \frac{\partial u}{\partial t} + i\frac{\partial^2 u}{\partial x^2} = 0$$

$$\text{The heat equation:} \quad \frac{\partial u}{\partial t} - a\frac{\partial^2 u}{\partial x^2} = 0$$

$$\text{The wave equation:} \quad \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0$$

Unless specified we assume $n = 2$.

3.1. The heat equation

We briefly discuss the heat equation before discussing solution methods. We write the three-dimensional heat equation as

$$\frac{\partial u}{\partial t} = D\,\Delta u$$

This equation describes the evolution of some temperature distribution $u \equiv u(\boldsymbol{r}, t)$. We call D the diffusion constant and have defined the Laplacian $\Delta \equiv \nabla^2$.

Proof. We consider heat diffusing through a material with conductivity k, specific heat capacity s, and density ρ. Let the heat inside a volume V be $Q(t)$. The heat flux through V is then

$$-\frac{\mathrm{d}Q}{\mathrm{d}t} = \int_{\partial V} (-k\boldsymbol{\nabla} u) \cdot \boldsymbol{n} \, \mathrm{d}A$$

which follows by definition. By the divergence theorem we can rewrite the RHS

$$-\frac{\mathrm{d}Q}{\mathrm{d}t} = \int_{V} \boldsymbol{\nabla} \cdot (-k\boldsymbol{\nabla} u) \, \mathrm{d}V$$

We can also write Q as

$$Q = \int s\rho u \, \mathrm{d}V$$

implying

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \int s\rho \frac{\partial u}{\partial t} \, \mathrm{d}V$$

Comparing the above immediately gives the heat equation. □

To make life simple we consider the one-dimensional heat equation

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0$$

We immediately see some trivial solutions

$$u(x,t) = x; \quad u(x,t) = \alpha x^2 + \beta x + \gamma + 2\alpha t$$

For non-trivial solutions we need $t > 0$ so u is defined on $\mathbb{R} \times (0, \infty)$. An important such solution is

$$u(x,t) = \frac{1}{\sqrt{4\pi t}} e^{-x^2/4t}$$

At a fixed time t_0 we define $\alpha_0 \equiv \sqrt{2t_0}$. The solution then takes the form

$$u(x,t_0) = \frac{1}{\sqrt{2\pi}\alpha_0} e^{-x^2/2\alpha_0^2}$$

which is Gaussian! And since $\alpha(t) = \sqrt{2t}$ the width of this Gaussian increases with time while the magnitude decreases.

3.2. First order PDEs

3.2.1. Linear homogeneous PDEs

We define the first-order linear *partial differential operator* $\mathcal{P}$ by

$$\mathcal{P} = A(x,y)\frac{\partial}{\partial x} + B(x,y)\frac{\partial}{\partial y} + C(x,y)$$

$$\mathcal{P} : C^1(X) \to C^0(X)$$

with $C^1(X)$ being the space of differential functions on $X \subset \mathbb{R}^2$. This guy acts on functions $u \in C^1(X)$ as

$$u(x,y) \mapsto (\mathcal{P}u)(x,y) = A(x,y)\frac{\partial u}{\partial x} + B(x,y)\frac{\partial u}{\partial y} + C(x,y)u$$

We say a partial differential equation is homogeneous if

$$\mathcal{P}u = 0$$

and it is linear if $\mathcal{P}$ is linear

$$\mathcal{P}(\alpha u + \beta v) = \alpha\mathcal{P}u + \beta\mathcal{P}v$$

Then by definition to solve a homogeneous partial differential equation we need to determine the kernel of $\mathcal{P}$.

3.2.2. Using equipotential lines

We assume $C = 0$. We will attempt to solve the problem by rewriting it in terms of an ordinary differential equation. When $n = 2$ we try to find equipotential lines defined by $p = p(x,y)$ along which $u(x,y)$ is constant meaning we can write $u(x,y) = f(p)$.

With this assumption we have

$$\frac{\partial u}{\partial x} = \frac{\mathrm{d}f}{\mathrm{d}p}\frac{\partial p}{\partial x}; \quad \frac{\partial u}{\partial y} = \frac{\mathrm{d}f}{\mathrm{d}p}\frac{\partial p}{\partial y}$$

implying

$$\underbrace{\left(A\frac{\partial p}{\partial x} + B\frac{\partial p}{\partial y}\right)}_{\overset{!}{=}0}\frac{\mathrm{d}f}{\mathrm{d}p} = 0$$

By definition of p we have

$$\mathrm{d}p = \frac{\partial p}{\partial x}\,\mathrm{d}x + \frac{\partial p}{\partial y}\,\mathrm{d}y = 0$$

implying

$$\frac{\mathrm{d}x}{A} = \frac{\mathrm{d}y}{B} \sim \text{parametrisation of lines}$$

Example 3.2. Consider

$$x\frac{\partial u}{\partial x} - 2y\frac{\partial u}{\partial y} = 0$$

Then

$$\frac{\mathrm{d}x}{x} = -\frac{\mathrm{d}y}{2y}$$

implies

$$x^2y = \text{constant} \overset{!}{=} p$$

Then the solution is

$$u(x,y) = f(x^2y)$$

with f being arbitrary.

Additional boundary conditions restrict our solution. Assuming

$$u(x=1,y) = 2y+1$$

we immediately find the particular solution

$$u(x,y) = 2x^2y+1$$

Assuming

$$u(x=1,y=1) = 4$$

we find many more solutions

$$u(x,y) = 4x^2y$$

$$= 3x^2y + 1$$

$$= 4$$

This happens since we are much less restrictive with this particular boundary condition.

When $C \neq 0$ the above does not work. We make the ansatz $u(x,y) = h(x,y)f(p)$ with $f(p)$ as above and $h(x,y)$ being arbitrary.

Example 3.3. Consider

$$x\frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial y} - 2u = 0$$

Using the ansatz we find

$$\left(x\frac{\partial h}{\partial x} + 2\frac{\partial h}{\partial y} - 2h\right)f + \left(x\frac{\partial p}{\partial x} + 2\frac{\partial p}{\partial y}\right)h\frac{\mathrm{d}f}{\mathrm{d}p} = 0$$

We assume we know some particular solution $h(x,y)$ to the original equation. Then the first term vanishes and we have

$$x\frac{\partial p}{\partial x} + 2\frac{\partial p}{\partial y} = 0$$

which we can solve as before! We find

$$p = xe^{-y/2}$$

and the general solution becomes

$$u(x,y) = h(x,y)f\big(xe^{-y/2}\big)$$

where $h(x,y)$ is any particular solution. An example in the above could be $h(x,y) = x^2$ which is easily seen by inspection.

3.2.3. Linear inhomogeneous PDEs

A partial differential equation is homogenoeus if given a solution $u(x,y)$ then $\lambda u(x,y)$ with $\lambda \in \mathbb{C}$ is also a solution. Clearly this is equivalent to the linear case where we had

$$\mathcal{P}u = 0$$

A partial differential boundary value problem is homogeneous if the partial differential equation is homogeneous and given that if $u(x, y)$ satisfies the boundary conditions then $\lambda u(x, y)$ satisfies the boundary conditions.

Example 3.4. An example of an inhomogeneous boundary value problem is

$$u(x, y) = 2x^2 y + 1$$

with $u(x = 1, y) = 2y + 1$ which we derived above.

With $v \in C^0(X)$ with $X \subset \mathbb{R}^2$ then an inhomogeneous partial differential equation can be written as

$$\mathcal{P}u = v$$

with $u \in C^1(X)$. This can also be written as

$$A(x, y)\frac{\partial u}{\partial x} + B(x, y)\frac{\partial u}{\partial y} + C(x, y)u + R(x, y) = 0$$

The solution of these equations are related to their homogeneous counterpart since the general solution can be written as

$$\text{particular solution to inhomogeneous} + \text{general solution to homogeneous}$$

Example 3.5. Consider

$$y\frac{\partial u}{\partial x} - x\frac{\partial u}{\partial y} = 3x$$

with $u(x, y = 0) = x^2$. We can find the solution to the homogeneous equation by equipotential lines

$$\frac{\mathrm{d}x}{y} = -\frac{\mathrm{d}y}{x} \Rightarrow p = x^2 + y^2$$

Then $u_{\text{homogeneous}}(x, y) = f(x^2 + y^2)$. We now need some particular solution to the inhomogeneous equation. By inspectiong it is easy to find $u_{\text{particular}}(x, y) = -3y$. The general solution is then

$$u(x, y) = f(x^2 + y^2) - 3y$$

Applying the boundary condition $u(x, y=0) = f(x^2) \stackrel{!}{=} x^2$ meaning $f(p) = p$ so

$$u(x,y) = x^2 + y^2 - 3y$$

3.2.4. Characteristics for first order PDEs

Consider the general first-order partial differential equation

$$A(x,y)\frac{\partial U}{\partial x} + B(x,y)\frac{\partial u}{\partial y} = F(x,y,u)$$

with the boundary condition $u(x,y) = \phi(s)$ which is defined along some curve $\Gamma \subset \mathbb{R}^2$. Let the curve be parametrised by $(x,y) = (x(s), y(s))$. Then

$$\frac{\mathrm{d}u}{\mathrm{d}s} = \frac{\partial u}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{\partial u}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}s}$$

These can be written as a system of equations in the following way

$$\underbrace{\begin{pmatrix} \frac{\mathrm{d}x}{\mathrm{d}s} & \frac{\mathrm{d}y}{\mathrm{d}s} \\ A(x,y) & B(x,y) \end{pmatrix}}_{M} \begin{pmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\mathrm{d}\phi}{\mathrm{d}s} \\ F(x,y,u) \end{pmatrix}$$

Given $\det M \neq 0$ this can be solved to determine $\boldsymbol{\nabla} u(x,y)$.

The condition $\det M = 0$ determines a set of curves we call the *characteristics*. This determinant leads to

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{B(x,y)}{A(x,y)}$$

which the curves must satisfy. This condition is the same which determined the equipotential lines.

When $\left(\frac{\mathrm{d}x}{\mathrm{d}s}, \frac{\mathrm{d}y}{\mathrm{d}s}\right)$ depends linearly on (A, B) we see that $\det M = 0$. This is equivalent to saying that the curve on which the boundary condition is specified coincides with the characteristics. Since these correspond to equipotential lines the function $u(x,y)$ is constant along the characteristics. Given this is not the case we can solve to find $\boldsymbol{\nabla} u$ and given $u(x,y) = \phi(s)$ on Γ we can specify the solution locally. This is just done by a Taylor expansion.

Example 3.6. Consider

$$x\frac{\partial u}{\partial x} - 2y\frac{\partial u}{\partial y} = 0$$

with $u(x, y) = 2y + 1$ on the line $x = 1$, $y \in [0, 1]$. We have already found the general solution to be

$$u(x, y) = f(x^2 y)$$

with the characteristics $c = x^2 y$. The particular solution obeying the boundary condition is

$$u(x, y) = 2x^2 y + 1$$

with $y \in [0, 1]$.

This solution is determined everywhere along a given characteristic since $f(x^2 y)$ is constant. The more general solution is

$$u(x, y) = 2x^2 y + 1 + g(x^2 y)$$

with $g(z) = 0$ for $0 \le z \le 1$.

Generally if Γ is itself a characteristic curve then the solution is unspecified everywhere expect on Γ. While if Γ crosses some characteristic multiple times then this can lead to overdetermination meaning no solution may exist. The line Γ should only cross each characteristic once and then the solution is specified along these.

3.3. Second order PDEs

The general second order partial differential equation is of the form

$$A(x, y)\frac{\partial^2 u}{\partial x^2} + B(x, y)\frac{\partial^2 u}{\partial y \partial x} + C(x, y)\frac{\partial^2 u}{\partial y^2} + D(x, y)\frac{\partial u}{\partial x} + E(x, y)\frac{\partial u}{\partial y} + F(x, y)u = R(x, y)$$

We classify these equations in the following way:

$$\text{hyperbolic} \longleftrightarrow B^2 > 4AC$$

$$\text{parabolic} \longleftrightarrow B^2 = 4AC$$

$$\text{elliptic} \longleftrightarrow B^2 < 4AC$$

This classification is local since the behaviour might change within $X \subseteq \mathbb{R}^2$.

3.3.1. Homogeneous second order PDEs with constant coefficients

We assume constant coefficients with $D = E = F = R = 0$. Then

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial y \partial x} + C\frac{\partial^2 u}{\partial y^2} = 0$$

We make the ansatz $u(x, y) = f(p)$ and obtain

$$A\left[\frac{\mathrm{d}^2 f}{\mathrm{d}p^2}\left(\frac{\partial p}{\partial x}\right)^2 + \frac{\mathrm{d}f}{\mathrm{d}p}\frac{\partial^2 p}{\partial x^2}\right] + B\left[\frac{\mathrm{d}^2 f}{\mathrm{d}p^2}\frac{\partial p}{\partial y}\frac{\partial p}{\partial x} + \frac{\mathrm{d}f}{\mathrm{d}p}\frac{\partial^2 p}{\partial y \partial x}\right] + C\left[\frac{\mathrm{d}^2 f}{\mathrm{d}p^2}\left(\frac{\partial p}{\partial y}\right)^2 + \frac{\mathrm{d}f}{\mathrm{d}p}\frac{\partial^2 p}{\partial y^2}\right] = 0$$

We assume p is linear in x and y. Then

$$p = ax + by$$

and we find

$$\underbrace{(Aa^2 + Bab + Cb^2)}_{\overset{!}{=}0}\frac{\mathrm{d}^2 f}{\mathrm{d}p^2} = 0$$

This is a quadratic

$$\lambda_{1,2} = \frac{b}{a} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2C}$$

Then any functions of

$$p_1 = x + \lambda_1 y; \quad p_2 = x + \lambda_2 y$$

satisfy the original equation. Then the general solution can be written as

$$u(x, y) = f(x + \lambda_1 y) + g(x + \lambda_2 y)$$

Example 3.7. Consider

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Then $A = C = 1$ and $B = 0$ meaning the equation is elliptic. We find $\lambda_{1,2} = \pm i$.

Example 3.8. Consider $A, C \in \mathbb{R}$ and $B = 0$. The solutions have arguments

$$\text{hyperbolic} \to x \pm \alpha y$$

$$\text{elliptic} \to x \pm i\beta y$$

with $\alpha, \beta \in \mathbb{R}$.

Example 3.9. Consider $B^2 = 4AC$. Then the solution is two-fold degenerate with

$$\lambda = -\frac{B}{2C}$$

yielding the solution

$$u(x,y) = f\left(x - \frac{B}{2C}y\right)$$

which is not completely general!

We would like to generalise our solution. We make the ansatz

$$u(x,y) = h(x,y)g\left(x - \frac{B}{2C}y\right)$$

and eventually obtain

$$\left(A\frac{\partial^2 h}{\partial x^2} + B\frac{\partial^2 h}{\partial y \partial x} + C\frac{\partial^2 h}{\partial y^2}\right)g = 0$$

Then h must be any solution to the original partial differential equation. The simplest such solution is just $h(x,y) = x$ giving the general solution for the parabolic case

$$u(x,y) = f\left(x - \frac{B}{2C}y\right) + xg\left(x - \frac{B}{2C}y\right)$$

which is quite nice.

A complementary derivation relies on using *variable transformations* since these can trivialise the partial differential equation. We start with the hyperbolic and elliptic cases where $B^2 \neq 4AC$. We define

$$\xi \equiv x + \lambda_1 y; \quad \eta \equiv x + \lambda_2 y$$

implying

$$\frac{\partial u}{\partial x} = \left(\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta}\right)u; \quad \frac{\partial u}{\partial y} = \left(\lambda_1 \frac{\partial}{\partial \xi} + \lambda_2 \frac{\partial}{\partial \eta}\right)u$$

Then the partial differential equation becomes

$$\left[A\left(\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta}\right)^2 + B\left(\frac{\partial}{\partial \xi} + \frac{\partial}{\partial \eta}\right)\left(\lambda_1 \frac{\partial}{\partial \xi} + \lambda_2 \frac{\partial}{\partial \eta}\right) + C\left(\lambda_1 \frac{\partial}{\partial \xi} + \lambda_2 \frac{\partial}{\partial \eta}\right)^2\right]u(\xi, \eta) = 0$$

We define $\lambda_{1,2}$ by

$$A + B\lambda_i + C\lambda_i^2 = 0$$

meaning they are the same as before. Then the partial differential equation becomes

$$[2A + B(\lambda_1 + \lambda_2) + 2C\lambda_1\lambda_2]\frac{\partial^2 u}{\partial \xi \partial \eta} = 0$$

Since $B^2 \neq 4AC$ the term in the brackets never vanishes so

$$\frac{\partial^2 u}{\partial \xi \partial \eta} \overset{!}{=} 0$$

This has the solution

$$\frac{\partial u}{\partial \eta} = F(\eta); \quad \frac{\partial u}{\partial \xi} = G(\xi)$$

implying

$$\begin{aligned} u(\xi, \eta) &= f(\xi) + g(\eta) \\ &= f(x + \lambda_1 y) + g(x + \lambda_2 y) \end{aligned}$$

as we found before.

For the parabolic case we assume $B^2 = 4AC$ and define

$$\lambda \equiv -\frac{B}{2C}$$

Then we do

$$\xi = x + \lambda y; \quad \eta = x$$

Which gives

$$A\frac{\partial^2 u}{\partial \eta^2} = 0$$

implying

$$u(x, y) = xg(x + \lambda y) + f(x + \lambda y)$$

These transformations are typically not easy to find.

3.3.2. Homogeneous mixed first and second order PDEs with constant coefficients

We consider partial differential equations of the form

$$\alpha \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$$

with $u \equiv u(x, t)$ and α constant. This is simply the heat equation.

The previous strategy of $u = f(p)$ will not work since we will not be able to factor $f(p)$. We could instead try separation of variables which we discuss later or try setting both sides equal to a constant. We instead proceed by dimensional analysis and define

$$\eta \equiv \frac{x^2}{\alpha t}$$

which is dimensionless. We then try $u = f(\eta)$. Then

$$\frac{\partial u}{\partial x} = \frac{2x}{\alpha t} \frac{\mathrm{d}f}{\mathrm{d}\eta}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{2}{\alpha t} \frac{\mathrm{d}f}{\mathrm{d}\eta} + \left(\frac{2x}{\alpha t}\right)^2 \frac{\mathrm{d}^2 f}{\mathrm{d}\eta^2}$$

$$\frac{\partial u}{\partial t} = -\frac{x^2}{\alpha t^2} \frac{\mathrm{d}f}{\mathrm{d}\eta}$$

meaning the partial differential equation becomes

$$4\eta \frac{\mathrm{d}^2 f}{\mathrm{d}\eta^2} + (2 + \eta) \frac{\mathrm{d}f}{\mathrm{d}\eta} = 0$$

which is an ordinary differential equation! We can solve this to find $\frac{\mathrm{d}f}{\mathrm{d}\eta}$ by separation of variables which can then be integrated to give

$$f(\eta) \propto \int_{\eta_0}^{\eta} \mu^{-1/2} e^{-\mu/4} \, \mathrm{d}\mu$$

$$\propto \int_{\zeta_0}^{\zeta = x/2\sqrt{\alpha t}} e^{-\nu^2}\,\mathrm{d}\nu$$

where we do $\zeta \equiv \eta^{1/2}/2$. Taking $\zeta_0 = 0$ this is the *error function*

$$u(x,t) \propto \operatorname{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

note that for $\zeta \geq \zeta_0$ we require $x \geq 0$ and $t > 0$.

3.3.3. Characteristics for second order PDEs

We consider a general linear second order partial differential equation of the form

$$A(x,y)\frac{\partial^2 u}{\partial x^2} + B(x,y)\frac{\partial^2 u}{\partial x \partial y} + C(x,y)\frac{\partial^2 u}{\partial y^2} = F\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right)$$

Generally the boundary conditions should be such that u and its first partial derivatives are specified along a suitable set of boundaries bordering or enclosing the region of interest.

We classify boundary conditions as follows:

1. *Dirichlet* boundary conditions specify the value of u at each point of the boundary.
2. *Neumann* boundary conditions specify the value of $\partial u/\partial n$ being the *normal derivative* at each point of the boundary. We define

$$\frac{\partial u}{\partial n} \equiv \boldsymbol{\nabla} u \cdot \hat{n}$$

with $\hat{n}$ being normal to the boundary at each point.

3. *Cauchy* boundary conditions specify u and $\partial u/\partial n$ at each point of the boundary.

We consider the third case in depth. Assume there is some curve $\Gamma \subseteq \mathbb{R}^2$ on which the boundary conditions are defined. We parametrise this cuve as $(x,y) = (x(s), y(s))$ where s is the *curve parameter*. We denote the boundary conditions along Γ by $u(x,y) = \phi(s)$ and $\partial u/\partial n = \psi(s)$. At each point of Γ we have

$$\mathrm{d}\boldsymbol{r} = \mathrm{d}x\hat{x} + \mathrm{d}y\hat{y} \sim \text{tangent}$$

$$\hat{n}\,\mathrm{d}s = \mathrm{d}y\hat{x} - \mathrm{d}x\hat{y} \sim \text{normal}$$

Then along Γ we have

$$\frac{\partial u}{\partial s} \equiv \boldsymbol{\nabla} u \cdot \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}s} = \frac{\partial u}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}s} + \frac{\partial u}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}s} = \frac{\mathrm{d}\phi(s)}{\mathrm{d}s}$$

$$\frac{\partial u}{\partial n} \equiv \boldsymbol{\nabla} u \cdot \hat{n} = \frac{\partial u}{\partial x}\frac{\mathrm{d}y}{\mathrm{d}s} - \frac{\partial u}{\partial y}\frac{\mathrm{d}x}{\mathrm{d}s} = \psi(s)$$

We can write this as

$$\begin{pmatrix} \frac{\mathrm{d}x}{\mathrm{d}s} & \frac{\mathrm{d}y}{\mathrm{d}s} \\ \frac{\mathrm{d}y}{\mathrm{d}s} & -\frac{\mathrm{d}x}{\mathrm{d}s} \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\mathrm{d}\phi}{\mathrm{d}s} \\ \psi \end{pmatrix}$$

These can be solved to find $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ along Γ which provide information about the local neighbourhood of Γ and can be used to determine u.

Consider

$$\frac{\mathrm{d}}{\mathrm{d}s} = \frac{\mathrm{d}x}{\mathrm{d}s}\frac{\partial}{\partial x} + \frac{\mathrm{d}y}{\mathrm{d}s}\frac{\partial}{\partial y}$$

giving

$$\frac{\mathrm{d}}{\mathrm{d}s}\frac{\partial u}{\partial x} = \frac{\mathrm{d}x}{\mathrm{d}s}\frac{\partial^2 u}{\partial x^2} + \frac{\mathrm{d}y}{\mathrm{d}s}\frac{\partial^2 u}{\partial x \partial y}$$

$$\frac{\mathrm{d}}{\mathrm{d}s}\frac{\partial u}{\partial y} = \frac{\mathrm{d}x}{\mathrm{d}s}\frac{\partial^2 u}{\partial y \partial x} + \frac{\mathrm{d}y}{\mathrm{d}s}\frac{\partial^2 u}{\partial y^2}$$

Then we have

$$\begin{pmatrix} A & B & C \\ \frac{\mathrm{d}x}{\mathrm{d}s} & \frac{\mathrm{d}y}{\mathrm{d}s} & 0 \\ 0 & \frac{\mathrm{d}x}{\mathrm{d}s} & \frac{\mathrm{d}y}{\mathrm{d}s} \end{pmatrix} \begin{pmatrix} \frac{\partial^2 u}{\partial x^2} \\ \frac{\partial^2 u}{\partial x \partial y} \\ \frac{\partial^2 u}{\partial y^2} \end{pmatrix} = \begin{pmatrix} F \\ \frac{\mathrm{d}}{\mathrm{d}s}\frac{\partial u}{\partial x} \\ \frac{\mathrm{d}}{\mathrm{d}s}\frac{\partial u}{\partial y} \end{pmatrix}$$

The characteristics are again determined by

$$\begin{vmatrix} A & B & C \\ \frac{\mathrm{d}x}{\mathrm{d}s} & \frac{\mathrm{d}y}{\mathrm{d}s} & 0 \\ 0 & \frac{\mathrm{d}x}{\mathrm{d}s} & \frac{\mathrm{d}y}{\mathrm{d}s} \end{vmatrix} = 0 \Rightarrow A\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 - B\frac{\mathrm{d}y}{\mathrm{d}x} + C = 0$$

This is an ordinary differential equation for the curves in the xy-plane along which the second partial derivatives of u cannot be found. These curves have

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{B \pm \sqrt{B^2 - 4AC}}{2A}$$

assuming $A \neq 0$. These can be classified as before $\sim$ hyperbolic, parabolic, elliptic.

Assuming A, B, C are constants then the characteristics are linear $x + \lambda y = \text{const}$.

Example 3.10. Consider

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0$$

We have $A = 1$, $C = -1/c^2$ and $B = 0$ meaning the equation is hyperbolic. Then

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \pm c$$

We find two families of characteristics

$$x \mp ct = \text{const}$$

The general solution is then given by

$$u(x, y) = \underbrace{f(x - ct)}_{\text{forward}} + \underbrace{g(x + ct)}_{\text{backward}}$$

We specify Cauchy boundary conditions u and $\partial u / \partial n$ along a segment on the x-axis i.e. $t = 0$ and $x \in [0, L]$ with $L > 0$. Then the solution is specified on every straight line in the family corresponding to a slope $+c$ and on every straight line in the family corresponding to a slope $-c$ which intersect on the line segment. Along the forward characteristics $p = x - ct \sim \text{constant}$ so $f(x - ct)$ is uniquely defined for the characteristics that intersect the line segment. Likewise for the backward characteristics $p = x + ct$. The solution is then *partially defined* when either $f(x - ct)$ or $g(x + ct)$ are fully known and fully defined when both are known.

Certain boundary conditions are appropriate if we want well-defined solutions for the various classes of equation:

hyperbolic with open boundary surface $\sim$ Cauchy

parabolic with open boundary surface $\sim$ Dirichlet or Neumann

$$\text{elliptic bounded by closed surface} \sim \text{Dirichlet or Neumann}$$

3.3.4. Uniqueness

We prove the uniqueness of the Poisson equation when using either Dirichlet or Neumann boundary conditions.

Theorem 3.11. Let u be real and let its first and second partial derivatives be continuous in a region $V \subseteq \mathbb{R}^n$ as well as on its boundary $S \equiv \partial V$. Furthermore, let

$$\nabla^2 u(\boldsymbol{r}) = \rho(\boldsymbol{r}) \text{ for } \boldsymbol{r} \in V$$

and either $u = f$ or $\partial u / \partial n = g$ on S. Then the solution u is unique.[1]

Proof. Let $u_1(\boldsymbol{r})$ and $u_2(\boldsymbol{r})$ be distinct solutions satisfying everything mentioned. We want to show that these differ by at most a constant. We define $\omega(\boldsymbol{r}) = u_1(\boldsymbol{r}) - u_2(\boldsymbol{r})$. Within V this function satisfies the Laplace equation

$$\nabla^2 \omega = 0$$

and on S we have

$$u_1 = u_2 = f \quad \text{or} \quad \frac{\partial u_1}{\partial n} = \frac{\partial u_2}{\partial n} = g$$

depending on which boundary conditions are given. Then we have

$$\omega = 0 \quad \text{or} \quad \frac{\partial \omega}{\partial n} = 0$$

on S. We now consider

$$\boldsymbol{\nabla} \cdot (\omega \boldsymbol{\nabla} \omega) = \omega \nabla^2 \omega + \boldsymbol{\nabla} \omega \cdot \boldsymbol{\nabla} \omega$$

integrating

$$\begin{aligned}\int_V \Bigg(\underbrace{\omega \nabla^2 \omega}_{0} + \boldsymbol{\nabla} \omega \cdot \boldsymbol{\nabla} \omega \Bigg) \mathrm{d}V &= \int_S \omega (\boldsymbol{\nabla} \omega \cdot \hat{n}) \, \mathrm{d}S \\ &= \int_S \omega \frac{\partial \omega}{\partial n} \, \mathrm{d}S\end{aligned}$$

[1] Up to some constant.

$$\overset{\text{by assumption}}{=} 0$$

We are left with

$$\int_V |\boldsymbol{\nabla}\omega|^2 \, \mathrm{d}V = 0 \Rightarrow \boldsymbol{\nabla}\omega = 0$$

implying $\omega = u_1 - u_2 \sim$ constant. $\square$

As a corollary it follows that for Dirichlet boundary conditions with $u_1 = u_2$ on S then $u_1 = u_2$ everywhere in V. While for Neumann boundary conditions u_1 and u_2 can differ by a constant in V.

4. Advanced Methods for PDEs

4.1. Separation of variables

We assume coordinates (x, y, z, t) and take our PDE to be defined on a set $X \subseteq \mathbb{R}^4$. We only consider homogeneous equations.

A solution is *separable* if we can factorise it as

$$u(x, y, z, t) = X(x)Y(y)Z(z)T(t)$$

We may refer to a solution as *partially separable* with the meaning being obvious.

Example 4.1 (The 3D wave equation)**.** We consider the equation

$$\nabla^2 u(\boldsymbol{r}) = \frac{1}{c^2}\frac{\partial^2 u(\boldsymbol{r})}{\partial t^2}$$

We make the ansatz $u = XYZT$ giving

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = \frac{1}{c^2}\frac{T''}{T}$$

This can only be true for all $\{x, y, z, t\}$ if each term is constant. We end up with four separable ODEs

$$\frac{X''}{X} = -l^2; \quad \frac{Y''}{Y} = -m^2; \quad \frac{Z''}{Z} = -n^2; \quad \frac{T''}{T} = -c^2\mu^2$$

along with the requirement

$$-(l^2 + m^2 + n^2) = -\mu^2$$

we refer to the constant as *separation constants*. All four ODEs have similar solutions with one being

$$X = Ae^{ilx} + Be^{-ilx}$$

where $A, B, ...$ are specified using boundary conditions.

Generally a separable solution with n independent variables has $n - 1$ independent separation constants.

Assuming the PDE is linear then the *superpositions principle* holds. Then we can form general solutions by constructing superpositions of solutions corresponding to different values of separation constants. Consider $X \subseteq \mathbb{R}^2$ with $u_{\lambda_1} = X_{\lambda_1} Y_{\lambda_1}$ being a solution where the separation constant takes the value λ_1. Then

$$u = \sum_i a_i X_{\lambda_i} Y_{\lambda_i}$$

is also a solution for any a_i provided the λ_i work.

4.2. Integral transforms

We show how the *Fourier transform*[2] can be used to solve PDEs by again considering the heat equation

$$\alpha \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad \text{for } t > 0 \text{ and } u(x,0) = f(x) \text{ with } x \in \mathbb{R}$$

To make life simple we assume $u \to 0$ and $\partial u \to 0$ as $|x| \to \infty$. Then we can take the Fourier transform of the PDE

$$\frac{\alpha}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{-ikx} \, \mathrm{d}x = \frac{1}{\sqrt{2\pi}} \frac{\partial}{\partial t} \int_{-\infty}^{\infty} u e^{-ikx} \, \mathrm{d}x$$

which by definition becomes

$$-\alpha k^2 \tilde{u}(k,t) = \frac{\partial \tilde{u}(k,t)}{\partial t}$$

with

$$\tilde{u} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u(x,t) e^{-ikx} \, \mathrm{d}x \sim \text{Fourier transform}$$

The solution is

$$\tilde{u}(k,t) = \tilde{u}(k,0) e^{-\alpha k^2 t}$$

where

$$\tilde{u}(k,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u(x,0) e^{-ikx} \, \mathrm{d}x = \tilde{f}(k)$$

We find

$$\tilde{u}(k,t) = \tilde{f}(k) e^{-\alpha k^2 t} = \sqrt{2\pi} \tilde{f}(k) \tilde{G}(k,t)$$

[2]Details in exercises.

where

$$\tilde{G}(k,t) = \frac{1}{\sqrt{2\pi}} e^{-\alpha k^2 t}$$

We see $\tilde{u}$ is a product of two Fourier transforms. We can then apply the *convolution theorem* to find

$$u(x,t) = \int_{-\infty}^{\infty} G(x-x',t) f(x') \, \mathrm{d}x'$$

We need $G(x,t)$[3]

$$G(x,t) \overset{\text{inverse Fourier}}{=} \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\alpha k^2 t} e^{ikx} \, \mathrm{d}k$$

$$= \frac{1}{\sqrt{4\pi\alpha t}} e^{-\frac{x^2}{4\alpha t}}$$

We have found the solution

$$u = \frac{1}{\sqrt{4\pi\alpha t}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-x')^2}{4\alpha t} \right) f(x') \, \mathrm{d}x'$$

Which becomes the previously discussed solution for $f(x) = \delta(x)$.

4.3. Green's functions

Let $\mathcal{P}$ denote a linear partial differential operator. Then a linear inhomogeneous PDE has the form

$$\mathcal{P} u(\boldsymbol{r}) = \rho(\boldsymbol{r})$$

where $\boldsymbol{r} \in \mathbb{R}^n$.

4.3.1. ODEs and Green's functions

Let $x, b \in \mathbb{C}^n$ and $A \in \mathrm{GL}(\mathbb{C}, n)$. We want to solve the linear system

$$Ax = b$$

By definition A is *invertible* so the solution is

$$x = A^{-1} b \equiv Mb$$

[3] This is the *Green's function* as defined below.

with $A_{ij}M_{jk} = \delta_{ik}$. We will see that finding a *Green's function* amounts to being the continuous equivalent of the matrix inverse.

Before defining the Green's function we define the *Dirac δ-function* by

$$\delta(x) = \lim_{\varepsilon \to 0^+} \delta_\varepsilon(x)$$

with

$$\delta_\varepsilon(x) = \frac{1}{\varepsilon\sqrt{\pi}} e^{-x^2/\varepsilon^2}$$

or more simply as

$$\delta(x) = \begin{cases} 0 & \text{for } x \neq 0 \\ \infty & \text{for } x = 0 \end{cases}$$

We can also think of it as the derivative of the *step function*. We care about the δ-function due to the property

$$\int \mathrm{d}x f(x)\delta(x - x_0) = f(x_0)$$

i.e. it picks out the value of f at x_0.

Consider a linear ODE of order n

$$\mathcal{P}y(x) = f(x)$$

where

$$\mathcal{P} = a_n(x)\frac{\mathrm{d}^n}{\mathrm{d}x^n} + a_{n-1}(x)\frac{\mathrm{d}^{n-1}}{\mathrm{d}x^{n-1}} + ... + a_1(x)\frac{\mathrm{d}}{\mathrm{d}x} + a_0(x)$$

We would like to invert $\mathcal{P}$ and write $y = \mathcal{P}^{-1}f$. We denote this object as the *Green's function*. Generalising the discrete $x_i = M_{ij}b_j$ we have

$$y(x) = \int_a^b G(x, z)f(z)\,\mathrm{d}z$$

Then

$$\mathcal{P}y(x) = \int_a^b \mathcal{P}G(x, z)f(z)\,\mathrm{d}z \overset{!}{=} f(x)$$

implying

$$\mathcal{P}G(x,z) = \delta(x-z)$$

with $a \le x \le b$. Then if we can find $G(x,z)$ we have found our solution!

Example 4.2. Consider the second order ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \delta(x-1)$$

with

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=0} = 0; \quad y(0) = 0$$

We consider two regions $x < 1$ and $x > 1$.

For $x < 1$ we have

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = 0 \Rightarrow y = Ax + B$$

with the boundary conditions we immediately have $A = B = 0$.

For $x > 1$ we have

$$y = Cx + D$$

However, now the boundary conditions are of no use. To proceed we show y is continuous at $x = 1$. By contradiction if it were discontinuous then $\frac{\mathrm{d}y}{\mathrm{d}x} \sim \delta(x-1)$ and

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} \sim \frac{\mathrm{d}}{\mathrm{d}x}\delta(x-1)$$

which is a contradiction. We can also show that $\frac{\mathrm{d}y}{\mathrm{d}x}$ is discontinuous as $x = 1$ since

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \Theta(x-1) + \mathrm{const} \sim \text{discontinuous}$$

By continuity we have

$$0 = Cx + D|_{x=1} \Rightarrow C = -D$$

By discontinuity we have

$$\underbrace{\lim_{x\to 1^-} \frac{\mathrm{d}y}{\mathrm{d}x}}_{0} + 1 = \underbrace{\lim_{x\to 1^+} \frac{\mathrm{d}y}{\mathrm{d}x}}_{C}$$

implying $C = 1$ and $D = -1$. Then

$$y(x) = \begin{cases} x-1 & \text{for } x \geq 1 \\ 0 & \text{for } x \leq 1 \end{cases}$$

$$= (x-1)\Theta(x-1)$$

We now consider more general properties of Green's functions for ODEs. We have

$$\sum_{m=0}^{n} a_m \frac{\mathrm{d}^m}{\mathrm{d}x^m} G(x,z) = 0; \text{ for } x < z \text{ and } x > z$$

Then

$$\lim_{\varepsilon \to 0} \int_{z-\varepsilon}^{z+\varepsilon} \mathcal{P} G(x,z)\,\mathrm{d}x = \lim_{\varepsilon \to 0} \sum_{m=0}^{n} \int_{z-\varepsilon}^{z+\varepsilon} a_m(x) \frac{\mathrm{d}^m}{\mathrm{d}x^m} G(x,z)\,\mathrm{d}x$$

$$= \lim_{\varepsilon \to 0} \int_{z-\varepsilon}^{z+\varepsilon} \delta(x-z)\,\mathrm{d}x$$

$$= 1$$

We can simplify the LHS since all

$$\frac{\mathrm{d}^m}{\mathrm{d}x^m} G(x,z) \text{ for } 0 \leq m < n-1 \sim \text{continuous}$$

by an argument similar to that in the previous example. These terms then vanish in the limit $\varepsilon \to 0$. The $(n-1)$th derivative has a finite discontinuity at $x = z$ and also vanishes. We are left with

$$1 = \lim_{\varepsilon \to 0} \int_{z-\varepsilon}^{z+\varepsilon} a_n(x) \frac{\mathrm{d}^n}{\mathrm{d}x^n} G(x,z)\,\mathrm{d}x$$

$$\overset{\text{ibp}}{=} \lim_{\varepsilon \to 0} \left[a_n(x) \frac{\mathrm{d}^{n-1}}{\mathrm{d}x^{n-1}} G(x,z) \right]_{z-\varepsilon}^{z+\varepsilon}$$

implying that $\frac{\mathrm{d}^{n-1}}{\mathrm{d}x^{n-1}} G(x,z)$ has a discontinuity of size a_n^{-1} at $x = z$. We have found n constraints being that G and all derivative up to order $n-2$ are continuous along with the above finite discontinuity.

4.3.2. PDEs and Green's functions

We consider the *Poisson equation*

$$\nabla^2 u(\boldsymbol{r}) = \rho(\boldsymbol{r})$$

and will use Green's functions to solve boundary value problems in a given volume $V \subseteq \mathbb{R}^n$ with inhomogeneous boundary conditions.[4]

We recall *Green's second identity*. Let $\phi(\boldsymbol{r})$ and $\psi(\boldsymbol{r})$ be scalar functions defined in some volume $V \subseteq \mathbb{R}^n$ with boundary $\mathcal{S} = \partial V$. Then[5]

$$\int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) \, \mathrm{d}V = \int_{\mathcal{S}} (\phi \boldsymbol{\nabla} \psi - \psi \boldsymbol{\nabla} \phi) \cdot \hat{n} \, \mathrm{d}\mathcal{S}$$

We need the G satisfying

$$\nabla^2 G(\boldsymbol{r}, \boldsymbol{r}_0) = \delta(\boldsymbol{r} - \boldsymbol{r}_0)$$

with $\boldsymbol{r}_0 \in V$. We define $\phi(\boldsymbol{r}) = u(\boldsymbol{r})$ and $\psi(\boldsymbol{r}) = G(\boldsymbol{r}, \boldsymbol{r}_0)$ and obtain

$$\int_V \left[u(\boldsymbol{r}) \underbrace{\nabla^2 G(\boldsymbol{r}, \boldsymbol{r}_0)}_{\delta(\boldsymbol{r}-\boldsymbol{r}_0)} - G(\boldsymbol{r}, \boldsymbol{r}_0) \underbrace{\nabla^2 u(\boldsymbol{r})}_{\rho(\boldsymbol{r})} \right] \mathrm{d}V_{\boldsymbol{r}} = \int_{\mathcal{S}} \left[u(\boldsymbol{r}) \frac{\partial G(\boldsymbol{r}, \boldsymbol{r}_0)}{\partial n} - G(\boldsymbol{r}, \boldsymbol{r}_0) \frac{\partial u(\boldsymbol{r})}{\partial n} \right] \mathrm{d}\mathcal{S}_{\boldsymbol{r}}$$

where we integrate over $\boldsymbol{r}$. We find

$$u(\boldsymbol{r}_0) = \int_V G(\boldsymbol{r}, \boldsymbol{r}_0) \rho(\boldsymbol{r}) \mathrm{d}V_{\boldsymbol{r}} + \int_{\mathcal{S}} \left[u(\boldsymbol{r}) \frac{\partial G(\boldsymbol{r}, \boldsymbol{r}_0)}{\partial n} - G(\boldsymbol{r}, \boldsymbol{r}_0) \frac{\partial u(\boldsymbol{r})}{\partial n} \right] \mathrm{d}\mathcal{S}_{\boldsymbol{r}}$$

which is our solution.

We assume Dirichlet boundary conditions meaning $u(\boldsymbol{r})$ is given on $\mathcal{S}$. Let $f(\boldsymbol{r}) : \mathbb{R}^n \to \mathbb{R}$ be a function with $u(\boldsymbol{r}) = f(\boldsymbol{r})$ for $\boldsymbol{r} \in \mathcal{S}$. We are free to choose the boundary condition satisfied by $G(\boldsymbol{r}, \boldsymbol{r}_0)$ so we pick $G(\boldsymbol{r}, \boldsymbol{r}_0) = 0$ for $\boldsymbol{r} \in \mathcal{S}$. This defines the *Dirichlet Green's function*. Then the third term on the RHS vanishes

$$u(\boldsymbol{r}_0) = \int_V G(\boldsymbol{r}, \boldsymbol{r}_0) \rho(\boldsymbol{r}) \mathrm{d}V_{\boldsymbol{r}} + \int_{\mathcal{S}} f(\boldsymbol{r}) \frac{\partial G(\boldsymbol{r}, \boldsymbol{r}_0)}{\partial n} \mathrm{d}\mathcal{S}_{\boldsymbol{r}}$$

We now need to find G satisfying $G(\boldsymbol{r}, \boldsymbol{r}_0) = 0$ for $\boldsymbol{r} \in \mathcal{S}$. We make the ansatz

[4] Dirichlet or Neumann. With both given the problem is *overdetermined*.

[5] This follows from the *divergence theorem* applied to $\boldsymbol{F} = \phi \boldsymbol{\nabla} \psi - \psi \boldsymbol{\nabla} \phi$.

$$G(\boldsymbol{r}, \boldsymbol{r}_0) = F(\boldsymbol{r}, \boldsymbol{r}_0) + H(\boldsymbol{r}, \boldsymbol{r}_0)$$

and require

$$\nabla^2 F = \delta(\boldsymbol{r} - \boldsymbol{r}_0); \quad \nabla^2 H = 0$$

within the volume V and $F + H = 0$ on $\mathcal{S}$. We refer to F as the *fundamental solution.*

Example 4.3. We consider the Poisson equation in three dimensions. We assume $F \to 0$ as $|\boldsymbol{r}| \to \infty$. We consider a sphere $\mathcal{S}$ of radius R centered at $\boldsymbol{r}_0$ and integrate over the enclosed volume V

$$\int_V \nabla^2 F \mathrm{d}V_{\boldsymbol{r}} = \underbrace{\int_V \delta(\boldsymbol{r} - \boldsymbol{r}_0) \mathrm{d}V_{\boldsymbol{r}}}_{1}$$

by Gauss'

$$\int_V \nabla^2 F \mathrm{d}V_{\boldsymbol{r}} = \int_{\mathcal{S}} \boldsymbol{\nabla} F \cdot \hat{n} \, \mathrm{d}\mathcal{S}$$

and by symmetry $F(\boldsymbol{r}, \boldsymbol{r}_0) = F(|\boldsymbol{r} - \boldsymbol{r}_0|) = F(r)$ implying F has the same value everywhere on $\mathcal{S}$. Writing the RHS in spherical coordinates we find[6]

$$\frac{\mathrm{d}F}{\mathrm{d}r} = \frac{1}{4\pi r^2} \Rightarrow F(r) = -\frac{1}{4\pi r} + K$$

since $F \to 0$ as $r \to \infty$ we have $K = 0$ and we obtain

$$F(\boldsymbol{r}, \boldsymbol{r}_0) = -\frac{1}{4\pi |\boldsymbol{r} - \boldsymbol{r}_0|}$$

Example 4.4. We can now solve

$$\nabla^2 u(\boldsymbol{r}) = -\frac{\rho(\boldsymbol{r})}{\varepsilon_0} \sim \text{Gauss' law}$$

in $V = \mathbb{R}^3$ with $u(\boldsymbol{r}) \to 0$ as $r \to \infty$.

[6]Using $\mathrm{d}\mathcal{S} = r^2 \sin\theta \, \mathrm{d}\theta \mathrm{d}\phi$ and $\boldsymbol{\nabla} f = \frac{\partial f}{\partial r} \hat{r}$.

The F found above fully specifies G since it already satisfies the boundary condition.[7] The surface integral in the solution we found vanishes aswell since $f(\boldsymbol{r}) = 0$. The solution is then

$$u(\boldsymbol{r}_0) = \int_V \frac{\rho(\boldsymbol{r})\mathrm{d}V_{\boldsymbol{r}}}{4\pi\varepsilon_0|\boldsymbol{r}-\boldsymbol{r}_0|}$$

which should be familiar! This example was made trivial since $V = \mathbb{R}^3$ meaning we did not need to find H.

We now consider more general $V \subset \mathbb{R}^3$ where $F \neq 0$ on $\mathcal{S}$. We will construct the corresponding H by the *method of images*. The idea is to add solutions H of the homogeneous equation to F as to cancel the contributions of F on $\mathcal{S}$ meaning $G(\boldsymbol{r}, \boldsymbol{r}_0) = 0$ on $\mathcal{S}$. We do this by adding *copies* of F outside the volume V[8] to F itself.

We have a single source $\delta(\boldsymbol{r}-\boldsymbol{r}_0)$ inside the volume V and add N image sources outside the volume V

$$\sum_l^N \underbrace{q_l}_{\text{strength}} \delta(\boldsymbol{r}-\boldsymbol{r}_l)$$

with $\boldsymbol{r}_l$ outside the volume V. With an image source outside the volume V its corresponding fundamental solution satisfies the homogeneous equation inside the volume V.[9] We can write

$$\begin{aligned} G(\boldsymbol{r}, \boldsymbol{r}_0) &= F(\boldsymbol{r}, \boldsymbol{r}_0) + \underbrace{\sum_l^N q_l F(\boldsymbol{r}, \boldsymbol{r}_l)}_{H(\boldsymbol{r}, \boldsymbol{r}_0)} \\ &= \text{real} + \text{images} \end{aligned}$$

We can then adjust $\boldsymbol{r}_l$ and q_l until $G(\boldsymbol{r}, \boldsymbol{r}_0) = 0$ on $\mathcal{S}$. With $G(\boldsymbol{r}, \boldsymbol{r}_0)$ we have the solution to the Dirichlet boundary value problem.

[7] We set $H = 0$.
[8] We call these *image sources*.
[9] Since $\delta(\boldsymbol{r}-\boldsymbol{r}_l) = 0$ everywhere inside the volume V.

Example 4.5. We consider

$$\nabla^2 u(\boldsymbol{r}) = 0 \text{ for } z > 0$$

with $u(\boldsymbol{r}) = f(x,y)$ for $z=0$. The volume is $V = \{(x,y,z) \in \mathbb{R}^3 : z > 0\}$ and the boundary is $\partial V = \mathbb{R}^2$ alongside the surface at infinity. The Green's function must satisfy $G(\boldsymbol{r},\boldsymbol{r}_0) = 0$ in the plane $z=0$ and $G(\boldsymbol{r},\boldsymbol{r}_0) \to 0$ as $|\boldsymbol{r}| \to \infty$. This is achieved by an image source at $\boldsymbol{r}_1$ being the reflection of $\boldsymbol{r}_0$ about $z=0$ with strength $q=-1$

$$G(\boldsymbol{r},\boldsymbol{r}_0) = -\frac{1}{4\pi|\boldsymbol{r}-\boldsymbol{r}_0|} + \frac{1}{4\pi|\boldsymbol{r}-\boldsymbol{r}_1|}$$

where $\boldsymbol{r}_0 = (x_0, y_0, z_0)$ and $\boldsymbol{r}_1 = (x_0, y_0, -z_0)$.

With $\rho(\boldsymbol{r}) = 0$ we find

$$u(\boldsymbol{r}_0) = \int_{\mathcal{S}} f(\boldsymbol{r}) \frac{\partial G(\boldsymbol{r},\boldsymbol{r}_0)}{\partial n} \mathrm{d}\mathcal{S}_{\boldsymbol{r}}$$

where the normal derivative in the xy-plane is

$$\frac{\partial G}{\partial n} = -\boldsymbol{\nabla} G \cdot \hat{z} = -\frac{\partial G}{\partial z}$$

We now assume Neumann boundary conditions[10]

$$f(\boldsymbol{r}) = \frac{\partial u(\boldsymbol{r})}{\partial n} \text{ on } \mathcal{S}$$

These satisfy a *self-consistency* relation

$$\int_{\mathcal{S}} f(\boldsymbol{r})\, \mathrm{d}\mathcal{S} = \int_{\mathcal{S}} \boldsymbol{\nabla} u(\boldsymbol{r}) \cdot \hat{n}\, \mathrm{d}\mathcal{S} \overset{\text{Gauss}}{=} \int_V \nabla^2 u(\boldsymbol{r})\, \mathrm{d}V \overset{\text{Poisson}}{=} \int_V \rho(\boldsymbol{r})\, \mathrm{d}V$$

Analogously to before we would be tempted to pick

$$\frac{\partial G(\boldsymbol{r},\boldsymbol{r}_0)}{\partial n} = 0$$

This is not allowed due to the self-consistency relation

$$\int_{\mathcal{S}} \frac{\partial G}{\partial n}\, \mathrm{d}\mathcal{S} = \int_{\mathcal{S}} \boldsymbol{\nabla} G \cdot \hat{n}\, \mathrm{d}\mathcal{S} = \int_V \nabla^2 G\, \mathrm{d}V = 1$$

[10]This uniquely determines a solution up to an additive constant as shown.

The simplest choice is instead

$$\frac{\partial G(\boldsymbol{r}, \boldsymbol{r}_0)}{\partial n} = \frac{1}{A_{\mathcal{S}}}$$

with $\boldsymbol{r}$ on $\mathcal{S}$ and $A_{\mathcal{S}}$ being the area of $\mathcal{S}$. This defines the *Neumann Green's function*. Then we find

$$u(\boldsymbol{r}_0) = \int_V G(\boldsymbol{r}, \boldsymbol{r}_0)\rho(\boldsymbol{r})\,\mathrm{d}V + \underbrace{\frac{1}{A}\int_{\mathcal{S}} u(\boldsymbol{r})\,\mathrm{d}\mathcal{S}}_{\langle u \rangle_{\mathcal{S}} \quad \sim \text{ constant}} - \int_{\mathcal{S}} G(\boldsymbol{r}, \boldsymbol{r}_0) f(\boldsymbol{r})\,\mathrm{d}\mathcal{S}$$

Where $\langle u \rangle_{\mathcal{S}}$ is a *freely specifiable* constant. Assuming $\mathcal{S}$ is at infinity and $u(\boldsymbol{r}) \to 0$ as $|\boldsymbol{r}| \to \infty$ we force $\langle u \rangle_{\mathcal{S}} = 0$. The construction of $G(\boldsymbol{r}, \boldsymbol{r}_0)$ proceeds as before by the method of images.[11] The *tuning* of q_l and $\boldsymbol{r}_l$ is now done to satisfy the *area condition*.

[11]Assuming $G(\boldsymbol{r}, \boldsymbol{r}_0) = F(\boldsymbol{r}, \boldsymbol{r}_0) + H(\boldsymbol{r}, \boldsymbol{r}_0)$ is motivated from $\nabla^2 G(\boldsymbol{r}, \boldsymbol{r}_0) = \delta(\boldsymbol{r} - \boldsymbol{r}_0)$ which defines $G(\boldsymbol{r}, \boldsymbol{r}_0)$.

5. Complex Analysis

We extend $\mathbb{R}$ to a *field* wherein the solution to

$$x^2 + 1 = 0$$

exists. To this end we define $i^2 \equiv -1$ and form expressions $z = a + ib \in \mathbb{C}$ with $a, b \in \mathbb{R}$.[12] We will assume basic knowledge of complex numbers.

5.1. Complex functions

Let $z = x + iy \in D \subseteq \mathbb{C}$. Then $f(z)$ is a *complex function* of the *complex variable* z if each $z \in D$ corresponds to one or more values $f(z)$. We write

$$f(z) = \underbrace{u(x, y)}_{\text{real}} + i\,\underbrace{v(x, y)}_{\text{imaginary}}$$

and will consider *single-valued* functions unless specified.

Let $D \subseteq \mathbb{C}$ be an open set with $z_0 \in D$ and $f : D \to \mathbb{C}$ be a function. We say f is *differentiable* in z_0 if and only if the limit

$$f'(z_0) \equiv \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \sim \text{complex derivative}$$

exists and is unique.[13] When f is differentiable for all $z_0 \in D$ we call f *holomorphic* on D. A single-valued and differentiable f on $D \subseteq \mathbb{C}$ is called *analytic* on D. When f is analytic on D expect at a finite number of points in D then these are called *singularities* of f.

Theorem 5.1 (Cauchy-Riemann). Let $D \subseteq \mathbb{C}$ be an open interval, $z_0 \in D$ and $f : D \to \mathbb{C}$ be a function. Let $z = x + iy$ and $z_0 = x_0 + iy_0$. Writing $f(x, y) = u(x, y) + iv(x, y)$ the following are equivalent:

1. f is complex differentiable in z_0.
2. f is *totally real differentiable* in z_0 and the *Cauchy-Riemann equations* hold

$$\frac{\partial u(x_0, y_0)}{\partial x} = \frac{\partial v(x_0, y_0)}{\partial y}; \quad \frac{\partial u(x_0, y_0)}{\partial y} = -\frac{\partial v(x_0, y_0)}{\partial x}$$

[12] Artihmetic operations are defined as usual and it is trivial to show that $\mathbb{C}$ forms a field.
[13] *Unique* meaning independent of how we take $\Delta z \to 0$.

Assuming f is differentiable in z_0 we immediately find

$$f'(z_0) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial f}{\partial x} \sim \text{along } x$$

$$if'(z_0) = i\frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} \sim \text{along } y$$

Where we use Cauchy-Riemann and uniqueness.

Consider the infinite series

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

with $a_n \in \mathbb{C}$. This is called a *power series* with coefficients a_n around z_0. We call the set of all z that converge the *domain of convergence*.

Theorem 5.2. To all $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ there exists *exactly* one $r \in [0, \infty) \cup \{\infty\}$ called the *radius of convergence* with the following properties:

1. in every *closed circle* $|z - z_0| \leq \rho < r$ the series is *uniformly absolutely convergent.*
2. for $|z - z_0| > r$ the series is *divergent.*
3. the *Cauchy-Hadamard formula* holds

$$\frac{1}{r} \equiv \lim_{n \to \infty} \sup \Big(\sqrt[n]{|a_n|} \Big)$$

Theorem 5.3. Let $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ be a power series with radius of convergence $r > 0$. Let the function $f(z)$ be specified by this series on $B_r(z_0)$[14] meaning for all $z \in B_r(z_0)$ we have

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

Then f is holomorphic on $B_r(z_0)$ and

[14] $B_r(z_0)$ is the *ball* of radius r around z_0.

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$$

for all $z \in B_r(z_0)$.

By induction we obtain for all $k \in \mathbb{N}$

$$f^{(k)}(z) = \sum_{n=0}^{\infty} \frac{(n+k)!}{n!} a_{n+k} (z - z_0)^n$$

for all $z \in B_r(z_0)$. When $z = z_0$ we find

$$f^{(k)}(z_0) = k! a_k \Rightarrow a_k = \frac{f^{(k)}(z_0)}{k!}$$

Which upon substituting gives us the *Taylor series*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^n$$

for all $z \in B_r(z_0)$.

5.2. Elementary functions

We can now redefine familiar functions in terms of complex variables.

Consider $z \in \mathbb{C}$ then we can write

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

which is the *complex exponential function*. We have the following properties:

1. The radius of convergence is $r = \infty$.
2. The derivative of e^z is e^z.
3. $e^{z+w} = e^z e^w$ for all $z, w \in \mathbb{C}$.
4. With $z = x + iy$ we have $e^z = e^x e^{iy}$ with $|e^z| = |e^x|$.

We also seek solutions $w \in \mathbb{C}$ to

$$e^w = z$$

The solution w is only defined modulo $2\pi i$. To see this we can write

$$z = re^{i\theta}$$

for $-\pi < \theta \leq \pi$. Then for any $k \in \mathbb{Z}$ there is an equivalent expression

$$z = re^{i(\theta+2\pi k)}$$

The *naive logarithm* is then

$$w = \ln z = \ln r + i(\theta + 2\pi k)$$

which is then infinitely multi-valued. The *principal value* of the logarithm is

$$\mathrm{Ln}(z) = \ln r + i\mathrm{Arg}(z)$$

$$= \ln r + i\theta$$

with $-\pi \leq \theta < \pi$. The above corresponds to introducing a *branch cut* on $\mathbb{R}^-$. We can now define the general *complex power*

$$t^z = e^{z \ln t}$$

for $t \in \mathbb{C} \setminus \{0\}$. When $z = n^{-1}$ for $n \in \mathbb{N}$ we have n distinct roots of t. This follows from

$$t^{1/n} = r^{1/n} \exp\left[i\frac{\theta + 2\pi k}{n}\right]$$

and the 2π-periodicity of e^z.

5.3. Singularities and zeros

Consider $D \subseteq \mathbb{C}$ open. When f is singular at $z = z_0$ and analytic for all $z \in D \setminus \{z_0\}$ then z_0 is called an *isolated singularity* of f.

When f has the form

$$f(z) = \frac{g(z)}{(z - z_0)^n}$$

with $n \in \mathbb{N}$ and $g(z)$ is analytic on a neighbourhood D with $z_0 \in D$ and $g(z_0) \neq 0$ then f has a *pole of order* n at $z = z_0$. When $n = 1$ we call the pole *simple*. Alternatively, we have

$$\lim_{z \to z_0} (z - z_0)^n f(z) = a$$

with $a \in \mathbb{C} \setminus \{0\}$. This definition has some consequences:

1. When a vanishes then z_0 is either a pole of order less than n or f is analytic at z_0.
2. When $a \to \infty$ the pole is of order higher than n.
3. When f has a pole at z_0 then $|f| \to \infty$ as z_0 is approached from any direction.
4. When no finite n yields a value a then z_0 is called an *essential singularity*.

To define the behaviour of $f(z)$ at *infinity* we need to specify what *infinity* means. We do this by considering

$$f(z) = f(\xi^{-1})$$

at $\xi = 0$.

Example 5.4. Consider $f(z) = e^z$ then in terms of ξ

$$f(\xi^{-1}) = \sum_{n=0}^{\infty} \frac{\xi^{-n}}{n!}$$

at $\xi = 0$ we have an essential singularity and therefore at $z = \infty$.

When $f(z_0) = 0$ we call z_0 a *zero* of f. These are classified analogously to poles, so if we can write

$$f(z) = (z - z_0)^n g(z)$$

with $n \in \mathbb{N}$ and $g(z_0) \neq 0$ then z_0 is called a *zero of order* n of f. When $n = 1$ we call the zero *simple*. When z_0 is a zero of order n of f then by definition it is a pole of order n of f^{-1}.

5.4. Complex integrals

When $f(z)$ is a complex single-valued continuous function in $D \subseteq \mathbb{C}$[15] then we can define the *complex integral* of $f(z)$ between two points A and B along some curve $\Gamma \subset D$. We consider a path Γ described by the parametrisation $z(t) = x(t) + iy(t)$ where $t \in [\alpha, \beta] \in \mathbb{R}$. Then along Γ we have

$$\mathcal{J} = \int_\Gamma f(z)\,\mathrm{d}z$$

or

$$\mathcal{J} = \int_\alpha^\beta u\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t - \int_\alpha^\beta v\frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t + i\int_\alpha^\beta u\frac{\mathrm{d}y}{\mathrm{d}t}\,\mathrm{d}t + i\int_\alpha^\beta v\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t$$

implying $\mathcal{J}$ exists when

[15] With D open.

$$\frac{\mathrm{d}x}{\mathrm{d}t} \text{ and } \frac{\mathrm{d}y}{\mathrm{d}t}$$

are continuous.

We note the *standard estimate*. Assume $|f(z)| \le M \in \mathbb{R}^+$ along Γ then

$$\left| \int_\Gamma f(z)\,\mathrm{d}z \right| \le \int_\Gamma |f(z)||\,\mathrm{d}z| \le M \int_\Gamma \mathrm{d}\Gamma = ML$$

with the length of Γ being L.

5.5. Cauchy's theorem

Theorem 5.5 (Cuachy's integral theorem)**.** Let $D \subseteq \mathbb{C}$ open, $f : D \to \mathbb{C}$ holomorphic and $f'(z)$ continuous at all points in and on a closed contour $\Gamma \subset D$. Then

$$\oint_\Gamma f(z)\,\mathrm{d}z = 0$$

Proof. Let $p, q : \mathbb{R}^2 \to \mathbb{R}$ be functions with continuous first derivatives in and on a closed contour $\Gamma \subset D$ bounding a domain $R \subset D$. Then

$$\iint_R \left(\frac{\partial p}{\partial x} + \frac{\partial q}{\partial y} \right) \mathrm{d}x\,\mathrm{d}y = \oint_\Gamma (p\,\mathrm{d}y - q\,\mathrm{d}x)$$

This is *Green's theorem* and follows from Gauss' thoerem with $\hat{n}\,\mathrm{d}s = \hat{x}\,\mathrm{d}y - \hat{y}\,\mathrm{d}y$. We also have

$$\begin{aligned} \mathcal{I} &= \oint_\Gamma (u\,\mathrm{d}x - v\,\mathrm{d}y) + i \oint_\Gamma (v\,\mathrm{d}x + u\,\mathrm{d}y) \\ &= -\iint_R \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \mathrm{d}x\,\mathrm{d}y + i \iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) \mathrm{d}x\,\mathrm{d}y \\ &\overset{\text{by Cauchy-Riemann}}{=} 0 \end{aligned}$$

Where the Cauchy-Riemann equations hold since f is assumed holomorphic. □

Consider now two closed contours in $\mathbb{C}$ denoted by Γ and γ where γ lies inside Γ. Then if f is analytic in the region between Γ and γ we have

$$\oint_\Gamma f(z)\,\mathrm{d}z = \oint_\gamma f(z)\,\mathrm{d}z$$

Proof. We imagine cutting open Γ and γ introducing two contours $\mathcal{C}_1$ and $\mathcal{C}_2$ connecting Γ and γ. We now have a closed *keyhole* contour $\mathcal{C}$ consisting of four pieces. By assumption f is holomorphic on $\mathcal{C}$ and the region it encloses to by Cauchy's theorem

$$\oint_{\mathcal{C}} f\,\mathrm{d}z = \int_\Gamma f\,\mathrm{d}z + \int_{\mathcal{C}_1} f\,\mathrm{d}z + \int_{\mathcal{C}_2} f\,\mathrm{d}z + \int_\gamma f\,\mathrm{d}z = 0$$

We integrate along $\mathcal{C}_1$ and $\mathcal{C}_2$ in opposite directions so in the limit where the introduced cut vanishes these contributions cancel. Likewise, we integrate along Γ and γ in opposite directions so

$$\oint_\Gamma f\,\mathrm{d}z = \oint_\gamma f\,\mathrm{d}z$$

□

5.6. Cauchy's integral formula

Theorem 5.6 (Cauchy's integral formula)**.** Let $D \subseteq \mathbb{C}$ open and $f : D \to \mathbb{C}$ holomorphic. Let $\Gamma \subset D$ be a closed contour and z_0 a point within Γ. Then

$$f(z_0) = \frac{1}{2\pi i} \oint_\Gamma \frac{f(z)}{z - z_0}\,\mathrm{d}z$$

Proof. Let γ be a circle of radius ρ around the point z_0 such that γ is completely inside Γ. Since f is holomorphic inside Γ the integrand

$$\frac{f(z)}{z - z_0}$$

is analytic in the region between Γ and γ. Then by the above corollary

$$\oint_\Gamma = \oint_\gamma$$

Any point z on γ is parametrised by

$$z = z_0 + \rho e^{i\theta}$$

so

$$\mathcal{J} = \oint_\gamma \frac{f(z)}{z - z_0}\,\mathrm{d}z$$

$$= i\int_0^{2\pi} f\big(z_0 + \rho e^{i\theta}\big)$$

We can take the limit $\rho \to 0$ so

$$\mathcal{J} = 2\pi i f(z_0)$$

and the result follows. □

Then the value of any analytic function inside Γ is fully determined by the values on Γ!

Theorem 5.7. Let $D \subseteq \mathbb{C}$ open and $f : D \to \mathbb{C}$ holomorphic. Let $\Gamma \subset D$ as above. When f is arbitrarily often complex differentiable on D then

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_\Gamma \frac{f(z)}{(z - z_0)^{n+1}}\,\mathrm{d}z$$

Proof. Trivial by induction. □

This allows us to determine the value of all derivatives within Γ by knowing the values on Γ.

Theorem 5.8 (Cauchy's inequality)**.** Let f be analytic inside and on a circle γ of radius R centered on the point $z_0 \in \mathbb{C}$. If $|f| \le M$ on γ with $M \in \mathbb{R}^+$ then

$$\left|f^{(n)}(z_0)\right| \le \frac{Mn!}{R^n}$$

Proof. We have

$$\left|f^{(n)}(z_0)\right| = \frac{n!}{2\pi}\left|\oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}}\,\mathrm{d}z\right|$$

Using $|z - z_0| = R$ and $L_\gamma = 2\pi R$ the standard estimate gives

$$\left|f^{(n)}(z_0)\right| \leq \frac{Mn!}{R^n}$$

□

We call a function f *entire* when it is holomorphic on $\mathbb{C}$.

Theorem 5.9 (Liouville's theorem)**.** An entire function which is bounded has to be a constant.

Proof. We consider arbitrary $z_0 \in \mathbb{C}$. Then by Cauchy's inequality

$$\left|f^{(1)}(z_0)\right| \leq \frac{M}{R}$$

as $R \to \infty$ we have $\left|f^{(1)}(z_0)\right| = 0$ implying $f^{(1)}(z_0) = 0$ since M exists. Given f is an entire function and z_0 is arbitrary then $f^{(1)}(z) = 0$ for all $z \in \mathbb{C}$ implying $f(z) \sim$ constant.

□

5.7. Taylor- and Laurent series

Theorem 5.10 (Taylor's theorem)**.** Let f be analytic inside and on a circle γ of radius R centered at the point $z = z_0$ with z in γ. Then

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n; \quad a_n = \frac{f^{(n)}(z_0)}{n!}$$

The *Taylor series* above is valid inside the region of analyticity and for any z_0 can be shown to be unique.

Proof. Because f is analytic inside γ the Cauchy integral formula is valid

$$f(z) = \frac{1}{2\pi i} \oint_\gamma \frac{f(\xi)}{\xi - z} \, \mathrm{d}\xi$$

Recall

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$$

when $|r| < 1$. Then

$$\frac{\xi - z_0}{\xi - z} = \frac{1}{1 - \frac{z-z_0}{\xi-z_0}}$$

$$= \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n$$

implying

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n$$

Then

$$f(z) = \frac{1}{2\pi i} \oint_\gamma \frac{f(\xi)}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n \mathrm{d}\xi$$

$$= \frac{1}{2\pi i} \sum_{n=0}^{\infty} (z - z_0)^n \oint_\gamma \frac{f(\xi)}{(\xi - z_0)^{n+1}} \mathrm{d}\xi$$

$$= \frac{1}{2\pi i} \sum_{n=0}^{\infty} (z - z_0)^n 2\pi i \frac{f^{(n)}(z_0)}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

and we are done. □

Theorem 5.11 (Identity theorem)**.** Let f and g be analytic in some region R and $f = g$ in $S \subset R$ containing an *accumulation point* z_0. Then $f = g$ in R.[16]

Proof. Consider the difference $h = f - g$. We want to show that if h vanishes on S then it vanishes on R. Let $z_0 \in S$ be an accumulation point. Then

[16]The identity theorem implies *analytic continuations* are unique.

$$h(z) = h(z_0) + h^{(1)}(z_0)(z - z_0) + \frac{1}{2}h^{(2)}(z_0)(z - z_0)^2 + ...$$

with $h^{(n)}(z_0) = f^{(n)}(z_0) - g^{(n)}(z_0)$. This converges on a circle γ extending to the boundary of R since h is analytic in R. Since $z_0 \in S$ is an accumulation point we have

$$h(z_0) = h^{(1)}(z_0) = h^{(2)}(z_0) = ... = 0$$

implying $h(z) = 0$ inside γ. We can repeat this by choosing a new arbitrary z_0 inside γ. We eventually find $h(z) = 0$ in R.

□

When f has a singularity inside γ at some $z = z_0$ we cannot write a Taylor series.

Theorem 5.12 (Laurent series)**.** Let f have a pole of order p at $z = z_0$ and be analytic at all other points inside and on γ. Then

$$f(z) = \underbrace{\frac{a_{-p}}{(z - z_0)^p} + ... + \frac{a_{-1}}{z - z_0}}_{\text{principal part}} + \underbrace{a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + ...}_{\text{analytic part}}$$

where $a_{-p} \neq 0$. This extension of the Taylor series is called a *Laurent series*.

Proof. By definition we can write $g(z) = (z - z_0)^p f(z)$ with $g(z_0) = b_0 \neq 0$. This function is analytic at $z = z_0$ so we can write a Taylor series

$$g(z) = \sum_{n=0}^{\infty} b_n (z - z_0)^n$$

implying

$$\begin{aligned} f(z) &= \frac{1}{(z - z_0)^p} \sum_{n=0}^{\infty} b_n (z - z_0)^n \\ &= \sum_{n=0}^{\infty} b_n (z - z_0)^{n-p} \\ &= \sum_{n=-p}^{\infty} a_n (z - z_0)^n \end{aligned}$$

where $b_n \equiv a_{n-p}$ and $a_{-p} = b_0 \neq 0$. $\square$

We have

$$b_n = \frac{g^{(n)}(z_0)}{n!}$$

$$= \frac{1}{2\pi i} \oint \frac{g(z)}{(z - z_0)^{n+1}} \, \mathrm{d}z$$

implying

$$a_n = \frac{1}{2\pi i} \oint \frac{g(z)}{(z - z_0)^{n+1+p}} \, \mathrm{d}z$$

$$= \frac{1}{2\pi i} \oint \frac{f(z)}{(z - z_0)^{n+1}} \, \mathrm{d}z$$

We can have cases where the series around z_0 has infinite terms

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

The principal part converges for $\mathbb{C} \setminus C_1$ defined by

$$|z - z_0| > R_{\text{pp}}$$

while the analytic part converges for C_2 defined by

$$|z - z_0| < R_{\text{ap}}$$

The Laurent series then converges within a punctured disk $D = \mathbb{C} \setminus C_1 \cap C_2$ when $R_{\text{ap}} > R_{\text{pp}}$. Then any f which is analytic on D can be expressed a Laurent series about z_0. When the principal part is finite C_1 contains a single point z_0 and C_2 could extend to infinity.

The Laurent series can be used to *classify* z_0. This is done as:

1. If f is analytic in $z = z_0$ then $a_n = 0$ for $n < 0$.
2. Let $a_n = 0$ for $n < 0$ and $a_0 = a_1 = ... = a_{m-1} = 0$. Then if for $m > 0$ we have $a_m (z - z_0)^m \neq 0$ then f has a zero of order m at z_0.

3. Let f be non-analytic at z_0 and assume there exists some $p \in \mathbb{N}$ such that $a_{-p} \neq 0$ and $a_{-p-k} = 0$ for all $k \in \mathbb{N}$. Then $f(z) = a_{-p}(z - z_0)^{-p} + ...$ has a pole of order p at $z = z_0$.
4. Let f be non-analytic at z_0 and assume there does not exist some $p \in \mathbb{N}$ such that $a_{-p} \neq 0$ and $a_{-p-k} = 0$ for all $k \in \mathbb{N}$. Then there are infinitely many negative powers of $(z - z_0)$ and z_0 is an essential singularity.

We call a_{-1} the *residue* of f at the pole z_0. We denote this by $\operatorname{Res}_{z=z_0} f(z)$ or $\operatorname{Res}(z_0)$.

Example 5.13. Consider

$$f(z) = \frac{1}{z(z-2)^3}$$

which has singularities at $z = 0$ and $z = 2$. We consider $z = 0$. Then

$$\begin{aligned} f(z) &= -\frac{1}{8z} \frac{1}{\left(1 - \frac{z}{2}\right)^3} \\ &\overset{\text{analytic at } z=0}{=} -\frac{1}{8z}\left[1 + (-3)\left(-\frac{z}{2}\right) + \frac{(-3)(-4)}{2!}\left(-\frac{z}{2}\right)^2 + ...\right] \\ &= -\frac{1}{8z} - \frac{3}{16} - \frac{3}{16}z - \frac{5}{32}z^2 + ... \end{aligned}$$

We see $z = 0$ is a pole of order $p = 1$ with residue $-\frac{1}{8}$.

Theorem 5.14. When f has a pole of order m then

$$\operatorname{Res}_{z=z_0} f(z) = \lim_{z \to z_0}\left[\frac{1}{(m-1)!} \frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}\left((z - z_0)^m f(z)\right)\right]$$

Proof. Consider the Laurent series around z_0

$$f(z) = \frac{a_{-m}}{(z - z_0)^m} + ... + \frac{a_{-1}}{z - z_0} + a_0 + a_1(z - z_0) + ...$$

Then

$$(z - z_0)^m f(z) = a_{-m} + a_{-m+1}(z - z_0) + ... + a_{-1}(z - z_0)^{m-1} + a_0(z - z_0)^m + ...$$

and

$$\frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}((z-z_0)^m f(z)) = (m-1)!a_{-1} + \sum_{n=1}^{\infty} b_n (z-z_0)^n$$

yielding the result upon taking $z \to z_0$. $\square$

When the pole is simple $m = 1$ the above reduces to

$$\mathrm{Res}_{z=z_0} f(z) = \lim_{z \to z_0} ((z-z_0)f(z))$$

We can simplify this further by assuming

$$f(z) = \frac{g(z)}{h(z)}$$

Then by *l'Hôpital's rule*

$$\mathrm{Res}_{z=z_0} f(z) = \frac{g(z_0)}{h^{(1)}(z_0)}$$

5.8. The residue theorem

We now consider integrals on a closed contour Γ which encircles points where f is non-analytic. Consider f has a pole of order m at $z = z_0$. Then

$$f(z) = \sum_{n=-m}^{\infty} a_n (z-z_0)^n$$

We seek to compute

$$\mathcal{I} = \oint_{\Gamma} f(z)\,\mathrm{d}z$$

with z_0 inside Γ and no other singularities of f. By Cauchy's integral theorem

$$\mathcal{I} = \oint_{\gamma} f(z)\,\mathrm{d}z$$

with γ being a circle inside Γ centered at z_0. We parametrise γ by $z = z_0 + \rho e^{i\theta}$ giving

$$\mathcal{I} = \sum_{n=-m}^{\infty} a_n \int_0^{2\pi} i\rho^{n+1} e^{i(n+1)\theta}\,\mathrm{d}\theta$$

We see all terms with $n \neq -1$ vanish so we obtain

$$\mathcal{J} = 2\pi i a_{-1}$$

implying the value of $\mathcal{J}$ is fully determined by the residue of f at z_0. The generalisation of the above is the *residue theorem*

Theorem 5.15 (Residue theorem)**.** Let f be continuous within and on a closed contour Γ and analytic, except for a finite number of poles within Γ. Then

$$\oint_\Gamma f(z)\,\mathrm{d}z = 2\pi i \sum_j \operatorname{Res}_{z=z_j} f(z)$$

Proof. The proof follows by the above observation. We can encircle each singularity and connect these in such a way that the internal contours cancel. □

Example 5.16 (Green's function for the wave equation)**.** Consider

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)u(\boldsymbol{r}, t) = f(\boldsymbol{r}, t)$$

We demand

$$\left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right)G(\boldsymbol{r}, t; \boldsymbol{r}', t') = \delta(\boldsymbol{r} - \boldsymbol{r}')\delta(t - t')$$

Then

$$u(\boldsymbol{r}, t) = \int G(\boldsymbol{r}, t; \boldsymbol{r}', t') f(\boldsymbol{r}', t')\,\mathrm{d}^3\boldsymbol{r}'\,\mathrm{d}t$$

We will now find G. We are interested in $t \geq t'$ and require for $t < t'$ then $G = 0$. This is referred to as the *retarded Green's function*. Consider

$$\delta(\boldsymbol{r} - \boldsymbol{r}')\delta(t - t') = \frac{1}{(2\pi)^4}\int \mathrm{d}^3k\,\mathrm{d}\omega e^{i\boldsymbol{k}\cdot(\boldsymbol{r}-\boldsymbol{r}')}e^{-i\omega(t-t')}$$

and

$$G(\boldsymbol{r}, t; \boldsymbol{r}', t') = \int \frac{\mathrm{d}^3k}{(2\pi)^{3/2}} \int \frac{\mathrm{d}\omega}{(2\pi)^{1/2}} \hat{G}(\boldsymbol{k}, \omega) e^{i\boldsymbol{k}\cdot(\boldsymbol{r}-\boldsymbol{r}')}e^{-i\omega(t-t')}$$

These reduce the problem to

$$\hat{G}(\boldsymbol{k},\omega)\left(\frac{\omega^2}{c^2}-\boldsymbol{k}^2\right)=\frac{1}{4\pi^2}$$

implying

$$\hat{G}(\boldsymbol{k},\omega)=-\frac{c^2}{4\pi^2}\frac{1}{k^2c^2-\omega^2}$$

Then

$$G(\boldsymbol{r},t;\boldsymbol{r}',t')=-\frac{c^2}{4\pi^2}\int\frac{\mathrm{d}^3k}{(2\pi)^{3/2}}\int\frac{\mathrm{d}\omega}{(2\pi)^{1/2}}\frac{1}{k^2c^2-\omega^2}e^{i\boldsymbol{k}\cdot(\boldsymbol{r}-\boldsymbol{r}')}e^{-i\omega(t-t')}$$

We need to compute

$$\mathcal{J}=\int_{-\infty}^{\infty}\mathrm{d}\omega\frac{e^{-i\omega\tau}}{k^2c^2-\omega^2}$$

where $\tau\equiv t-t'$. The integral $\mathcal{J}$ is over $\mathbb{R}$ and has poles at $\omega=\pm kc$ where $k=|\boldsymbol{k}|$. We wish to use the residue theorem so we apply the *$i\varepsilon$-prescription* shifting the poles to

$$\pm kc\to\pm kc-i\varepsilon$$

with ε small. This allows us to perform the integral along $\mathbb{R}$. We then enclose the poles by a semi-circle contour Γ with radius $\rho>kc$. The corresponding integral is

$$\mathcal{J}_\varepsilon=\oint_\Gamma\mathrm{d}\omega\frac{e^{-i\omega\tau}}{k^2c^2-(\omega+i\varepsilon)^2}$$

We would like $\mathcal{J}_\varepsilon\to\mathcal{J}$ as $\rho\to\infty$.[17] This happens if the contribution from the semi-circle vanishes. The semi-circle is parametrised by $\omega=\rho e^{i\varphi}$ with $\pi\leq\varphi<2\pi$ so in the $\varepsilon\to 0$ limit for simplicity we have

$$\left|\frac{e^{-i\omega\tau}}{k^2c^2-\omega^2}\right|=\left|\frac{e^{-i\rho(\cos\varphi+i\sin\varphi)\tau}}{k^2c^2-\rho^2(\cos 2\varphi+i\sin 2\varphi)}\right|$$

$$=\frac{e^{\rho\sin\varphi\tau}}{|k^2c^2-\rho^2(\cos 2\varphi+i\sin 2\varphi)|}$$

and $\left|\omega^2\right|=\rho^2$. Since $\sin\varphi\leq 0$ for $\pi\leq\varphi<2\pi$ and demanding $\tau\geq 0$ we find for $\rho\gg kc$

[17] This is somewhat trivial since $\omega\to -i\infty$ so we have exponential supression.

$$\left|\frac{e^{-i\omega\tau}}{k^2c^2-\omega^2}\right| \overset{\rho\to\infty}{\sim} \frac{e^{\rho\sin\varphi\tau}}{\rho^2}$$

implying

$$\left|\int_\gamma \mathrm{d}\omega \frac{e^{-i\omega\tau}}{k^2c^2-\omega^2}\right| \leq \int_\pi^{2\pi} \mathrm{d}\varphi\rho\frac{e^{\rho\sin\varphi\tau}}{|k^2c^2-\rho^2|}$$

$$\overset{\rho\to\infty}{\to} 0$$

Then $\mathcal{J}_\varepsilon = \mathcal{J}$. Then by the residue theorem[18]

$$\mathcal{J} = -2\pi i(\mathrm{Res}_{\omega=kc} + \mathrm{Res}_{\omega=-kc})$$

We have

$$\frac{e^{-i\omega\tau}}{k^2c^2-\omega^2} = \frac{e^{-i\omega\tau}}{(kc-\omega)(kc+\omega)}$$

implying

$$\mathrm{Res}_{\omega=kc} = \lim_{\omega\to kc}\frac{e^{-i\omega\tau}}{(kc-\omega)(kc+\omega)}(\omega-kc)$$

$$= -\frac{e^{-ikc\tau}}{2kc}$$

$$\mathrm{Res}_{\omega=-kc} = \lim_{\omega\to -kc}\frac{e^{-i\omega\tau}}{(kc-\omega)(kc+\omega)}(\omega+kc)$$

$$= \frac{e^{ikc\tau}}{2kc}$$

Then

$$\mathcal{J} = -\frac{\pi i}{kc}\left(e^{ikc\tau} - e^{-ikc\tau}\right)$$

$$= -\frac{\pi i}{kc}2i\sin kc\tau$$

$$= \frac{2\pi\sin kc\tau}{kc}$$

[18]We have an extra − sign since we close the contour in the lower-half plane.

The last thing required to determine G is then computing the k-space integral. This is annoying so we will not do it.